第四章：线性空间与线性变换

2.

解. 令 $a_1f_1+a_2f_2+a_3f_3=0$, 则 $a_1+a_2(x-1)+a_3(x-1)^2=0$

$$a_1-a_2+a_3+(a_2-2a_3)x+a_3x^2=0$$

因为 $1,x,x^2$ 线性无关，所以

$$\begin{cases} a_1-a_2+a_3=0 \\ a_2-2a_3=0 \\ a_3=0 \end{cases}$$

解得 $a_1=a_2=a_3=0$, 所以 f_1,f_2,f_3 线性无关. 而 $P_2[x]$ 的维数为 3，所以 f_1,f_2,f_3 是一组基。下面令

$$a_1-a_2+a_3+(a_2-2a_3)x+a_3x^2=5x^2+x+3$$

得

$$\begin{cases} a_1-a_2+a_3=3 \\ a_2-2a_3=1 \\ a_3=5 \end{cases}$$

解得

$$\begin{cases} a_1=9 \\ a_2=11 \\ a_3=5 \end{cases}$$

所以 $g(x)$ 在此基下的坐标:$[9,11,5]^T$. □

3.

解. (1)

$$\begin{bmatrix} 1 & 1+x & (1+x)^2 & (1+x)^3 \end{bmatrix} = \begin{bmatrix} 1 & x & x^2 & x^3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

所以过渡矩阵为 $M = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(2) $f(x)$ 在基 $1, x, x^2.x^3$ 下坐标为 $[a_0, a_1, a_2, a_3]^T$, 在基 $1, 1+x, (1+x)^2, (1+x)^3]$ 下坐标为

$$M^{-1} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_0 - a_1 + a_2 - a_3 \\ a_1 - 2a_2 + 3a_3 \\ a_2 - 3a_3 \\ a_3 \end{bmatrix}$$

□

6.

解. (1)

$$\begin{bmatrix} \xi_1 & \xi_2 & \xi_3 \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{bmatrix} M_1$$

$$\begin{bmatrix} \eta_1 & \eta_2 & \eta_3 \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{bmatrix} M_2$$

M_1, M_2 都可逆，且该线性空间维数为 3，所以 ξ_1, ξ_2, ξ_3 以及 η_1, η_2, η_3 都是一组基.

(2) 令

$$\begin{bmatrix} \eta_1 & \eta_2 & \eta_3 \end{bmatrix} = \begin{bmatrix} \xi_1 & \xi_2 & \xi_3 \end{bmatrix} A$$

$$\begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{bmatrix} M_2 = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{bmatrix} M_1 A$$

$$A = M_1^{-1} M_2 = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 3 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

□

7.

解. (1)

$$\begin{bmatrix} \beta_1 & \beta_2 & \beta_3 & \beta_4 \end{bmatrix} = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha_2 + \alpha_3 + \alpha_4 & \alpha_1 + \alpha_3 + \alpha_4 & \alpha_1 + \alpha_2 + \alpha_4 & \alpha_1 + \alpha_2 + \alpha_3 \end{bmatrix}$$

所以，另一组基为:

$$\beta_1 = \alpha_2 + \alpha_3 + \alpha_4 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\beta_2 = \alpha_1 + \alpha_3 + \alpha_4 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\beta_3 = \alpha_1 + \alpha_2 + \alpha_4 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\beta_4 = \alpha_1 + \alpha_2 + \alpha_3 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

(2)
矩阵在 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 下的坐标为 $(0, 1, 2, -3)^T$. 在 $\beta_1, \beta_2, \beta_3, \beta_4$ 下坐标为

$$A^{-1}(0, 1, 2, -3)^T = (0, -1, -2, 3)^T$$

□

8.

解. 令

$$e_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, e_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

则，e_1, e_2, e_3, e_4 为 $R^{2\times 2}$ 的一组基，且

$$\begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & e_3 & e_4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & e_3 & e_4 \end{bmatrix} M_1$$

$$\begin{bmatrix} \beta_1 & \beta_2 & \beta_3 & \beta_4 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & e_3 & e_4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & e_3 & e_4 \end{bmatrix} M_2$$

显然 M_1, M_2 都可逆，且该线性空间维数为 4，所以这两组向量都是基.
令 $\alpha = \begin{bmatrix} a_0 & a_1 \\ a_2 & a_3 \end{bmatrix}$
α 在 e_1, e_2, e_3, e_4 下的坐标为 $\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$
所以，α 在 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 下的坐标为

$$M_1^{-1} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_0 - a_1 \\ a_1 - a_2 \\ a_2 - a_3 \\ a_3 \end{bmatrix}$$

(2)

$$\begin{bmatrix} \beta_1 & \beta_2 & \beta_3 & \beta_4 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & e_3 & e_4 \end{bmatrix} M_2$$

$$= \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \end{bmatrix} M_1^{-1} M_2 = \begin{bmatrix} \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 \end{bmatrix} \begin{bmatrix} -2 & 2 & 0 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & -2 & 2 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

即，过渡矩阵为 $\begin{bmatrix} -2 & 2 & 0 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & -2 & 2 \\ 1 & 1 & 1 & -1 \end{bmatrix}$ □

9.

解. (1) 方程组通解为 $k_1(1,0,...,-1)^T+k_2(0,1,...,-1)^T+...+k_{n-1}(0,0,...,1,-1)$,
所以解空间的维数为 $n-1$, 基为 $(1,0,...,-1)^T,(0,1,...,-1)^T,...,(0,0,...,1,-1)$
(2) $A=\begin{bmatrix} 2 & -4 & 5 & 3 \\ 3 & -6 & 4 & 2 \\ 4 & -8 & 17 & 11 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & -1 & -1 \\ 2 & -4 & 5 & 3 \\ 4 & -8 & 17 & 11 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & -1 & -1 \\ 0 & 0 & 7 & 5 \\ 0 & 0 & 21 & 15 \end{bmatrix} \to$
$\begin{bmatrix} 1 & -2 & -1 & -1 \\ 0 & 0 & 7 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
得到方程组通解为 $k_1(2,1,0,0)^T+k_2(2,0,-5,7)^T$
所以解空间维数为 2, 基为 $(2,1,0,0)^T,(2,0,-5,7)^T$ □

10.

解.

$$\begin{bmatrix} \alpha_1 & \cdots & \alpha_{i-1} & \alpha_{i+1} & \cdots & \alpha_{n+1} \end{bmatrix} = \begin{bmatrix} \alpha_1 & \cdots & \alpha_n \end{bmatrix} \begin{bmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 & x_1 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 & x_{i-1} \\ 0 & \cdots & 0 & 0 & \cdots & 0 & x_i \\ 0 & \cdots & 0 & 1 & \cdots & 0 & x_{1+1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 1 & x_n \end{bmatrix}$$

该矩阵对应行列式值为 $(-1)^{n+i}x_i$, 不为 0, 即矩阵可逆, 所以 $\alpha_1,\cdots,\alpha_{i-1},\alpha_{i+1},\cdots,\alpha_{n+1}$ 线性无关，又线性空间为 n 维，所以，任取 n 个向量都是一组基.

$$\alpha_{n+1}=x_1\alpha_1+x_2\alpha_2+...+x_n\alpha_n$$
$$\frac{1}{x_1}\alpha_{n+1}=\alpha_1+\frac{x_2}{x_1}\alpha_2+...+\frac{x_n}{x_1}\alpha_n$$
$$\alpha_1=-\frac{x_2}{x_1}\alpha_2-...-\frac{x_n}{x_1}\alpha_n+\frac{1}{x_1}\alpha_{n+1}$$

所以 α_1 在这组基下坐标为 $[-\frac{x_2}{x_1},-\frac{x_3}{x_1},...,-\frac{x_n}{x_1},\frac{1}{x_1}]^T$ □

13.

解. $c_1c_3\neq 0$, 所以 $\alpha=-\frac{1}{c_1}(c_2\beta+c_3\gamma)$
$\forall v\in span(\alpha,\beta),\exists\lambda_1,\lambda_2\in R, v=\lambda_1\alpha+\lambda_2\beta$
$v=(\lambda_2-\frac{c_2\lambda_1}{c_1})\beta-\frac{c_3\lambda_1}{c_1}\gamma\in span(\beta,\gamma)$
$span(\alpha,\beta)\subseteq span(\beta,\gamma)$
同理可证 $span(\alpha,\beta)\supseteq span(\beta,\gamma)$, 得到 $span(\alpha,\beta)=span(\beta,\gamma)$
□

15.

解. (1).
充分性:
令 $B = [\beta_1, \beta_2, ..., \beta_n]$, 则 $C(B) = span(\beta_1, ..., \beta_n)$.
而 $C(B) \subseteq N(A)$.
所以 $\forall \beta_i \in C(B), \beta_i \in N(A)$
即 $A\beta_i = 0$, 也就是说 $A\beta_1 = 0, ..., A\beta_n = 0$.
所以 $AB = A[\beta_1, ..., \beta_n] = 0$.
必要性:
$AB = 0$, 则 $A[\beta_1, ..., \beta_n] = 0$
即 $\forall \beta_i \in C(B), A\beta_i = 0, \beta_i \in N(A)$
所以 $C(B) \subseteq N(A)$. 充要性得证.
(2).
由 (1) 得, $AB = 0 \Rightarrow C(B) \subseteq N(A)$
所以 $dim(C(B)) \leq dim(N(A))$
而 $N(A)$ 就是 $Ax = 0$ 的解空间, 所以 $dim(N(A)) = n - r(A)$
所以 $r(B) = dim(C(B)) \leq dim(N(A)) = n - r(A)$, 即 $r(A) + r(B) \leq n$ □

16.

解. $W_1 + W_2 = span(\alpha_1, \alpha_2, \beta_1, \beta_2)$

$$[\alpha_1, \alpha_2, \beta_1, \beta_2] = \begin{bmatrix} 2 & 0 & 1 & 1 \\ 0 & -2 & 1 & -3 \\ 1 & 1 & 0 & 2 \\ 3 & 5 & -1 & 0 \\ -1 & -3 & 1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

所以 $W_1 + W_2$ 的维数为 3, 一组基为 $\alpha_1, \alpha_2, \beta_2$.
根据维数公式 $dim(W_1 \cap W_2) = dimW_1 + dimW_2 - dim(W_1 + W_2) = 1$
注意到 $\beta_1 \in W_2, \beta_1 = \frac{1}{2}\alpha_1 - \frac{1}{2}\alpha_2 \in W_1$, 所以 $\beta_1 \in W_1 \cap W_2$, 即 β_1 可以作为 $W_1 \cap W_2$ 的一组基 □

18.

解. (1) 当 $A = A_1$ 时,
$< \alpha, \beta >= \alpha^T A_1 \beta = a_1 b_1 + 2a_2 b_2 + a_3 b_3 = \beta^T A_1 \alpha =< \beta, \alpha >$
$< k\alpha, \beta >= k\alpha^T A_1 \beta = k < \alpha, \beta >$
$< \alpha_1 + \alpha_2, \beta >= (\alpha_1 + \alpha_2)^T A_1 \beta = (\alpha_1^T + \alpha_2^T) A_1 \beta = \alpha_1^T A_1 \beta + \alpha_2^T A_1 \beta =< \alpha_1, \beta > + < \alpha_2, \beta >$
$< \alpha, \alpha >= a_1^2 + 2a_2^2 + a_3^2 \geq 0$ 恒成立, 且当且仅当 $\alpha = 0$ 时, $< \alpha, \alpha >= 0$
所以, 当 $A = A_1$ 时为内积.
(2) 当 $A = A_2$ 时,
$< \alpha, \alpha >= a_1^2 + a_2^2 - 2a_1 a_2 + 2a_1 a_3 = (a_1 - a_2)^2 + 2a_1 a_3$
取 $a_1 = a_2 > 0, a_3 < 0$, 则 $< \alpha, \alpha >< 0$.
所以, 当 $A = A_2$ 时不为内积.
(3) 当 $A = A_3$ 时,
$< \alpha, \beta >= a_1 b_1 - a_2 b_1 + a_3 b_1 - a_1 b_2 + 2a_2 b_2 + 3a_1 b_3 + 3a_3 b_3$
$< \beta, \alpha >= a_1 b_1 - a_2 b_1 + 3a_3 b_1 - a_1 b_2 + 2a_2 b_2 + a_1 b_3 + 3a_3 b_3$
$< \alpha, \beta > \neq < \beta, \alpha >$
所以, 当 $A = A_3$ 时不为内积.

□

19.

证. (1)$< \beta, \alpha >= b_1a_1 - b_2a_1 - b_1a_2 + 3b_2a_2 = a_1b_1 - a_2b_1 - a_1b_2 + 3a_2b_2 =< \alpha, \beta >$
(2)$< k\alpha, \beta >= ka_1b_1 - ka_2b_1 - ka_1b_2 + 3ka_2b_2 = k(a_1b_1 - a_2b_1 - a_1b_2 + 3a_2b_2) = k < \alpha, \beta >$
(3) 取 $\gamma = [c_1, c_2]^T$, $< \alpha + \beta, \gamma >= (a_1 + b_1)c_1 - (a_2 + b_2)c_1 - (a_1 + b_1)c_2 + 3(a_2 + b_2)c_2 = a_1c_1 - a_2c_1 - a_1c_2 + 3a_2c_2 + b_1c_1 - b_2c_1 - b_1c_2 + 3b_2c_2 =< \alpha, \gamma > + < \beta, \gamma >$
(4)$< \alpha, \alpha >= a_1^2 - a_2a_1 - a_1a_2 + 3a_2^2 = a_1^2 - 2a_1a_2 + 3a_2^2 = (a_1 - a_2)^2 + 2a_2^2 \geq 0$, 当且仅当 $a_1 = a_2 = 0$ 时 $< \alpha, \alpha >= 0$.
综上，这是 R^2 的一个内积. □

21.

证. 假设 $\beta_1, \beta_2, \alpha_1, \alpha_2, ..., \alpha_r$ 线性相关，则存在不全为 0 的数 $l_1, l_2, k_1, k_2, ..., k_r$, 使得

$$l_1\beta_1 + l_2\beta_2 + k_1\alpha_1 + k_2\alpha_2 + ... + k_r\alpha_r = 0$$

首先 l_1, l_2 不全为 0，否则 $\alpha_1, ..., \alpha_r$ 线性相关，矛盾.
在等式两边与 $l_1\beta_1 + l_2\beta_2$ 做内积. 得

$$< l_1\beta_1 + l_2\beta_2, l_1\beta_1 + l_2\beta_2 + k_1\alpha_1 + k_2\alpha_2 + ... + k_r\alpha_r >= 0$$

$$< l_1\beta_1 + l_2\beta_2, l_1\beta_1 + l_2\beta_2 >= 0$$

$$l_1\beta_1 + l_2\beta_2 = 0$$

所以 β_1, β_2 线性相关，矛盾.
所以，$\beta_1, \beta_2, \alpha_1, \alpha_2, ..., \alpha_r$ 线性无关. □

22.

证. α_1, α_2 已经正交，只需标准化.

$$q_1 = \frac{\alpha_1}{||\alpha_1||} = \frac{1}{\sqrt{2}}[1, 0, 1, 0]^T$$

$$q_2 = \frac{\alpha_2}{||\alpha_2||} = \frac{1}{\sqrt{5}}[0, 1, 0, 2]^T$$

设 $\beta = [x_1, x_2, x_3, x_4]^T \in R^4$, 且 $\beta \perp \alpha_1, \beta \perp \alpha_2$
得到

$$\begin{cases} x_1 + x_3 = 0 \\ x_2 + 2x_4 = 0 \end{cases}$$

得到基础解系为

$$\alpha_3 = [-1, 0, 1, 0]^T, \alpha_4 = [0, -2, 0, 1]^T$$

已正交，只需标准化.

$$q_3 = \frac{\alpha_3}{||\alpha_3||} = \frac{1}{\sqrt{2}}[-1, 0, 1, 0]^T$$

$$q_4 = \frac{\alpha_4}{||\alpha_4||} = \frac{1}{\sqrt{5}}[0, -2, 0, 1]^T$$

所以 q_1, q_2, q_3, q_4 为 R^4 的一组标准正交基. □

24.

证. (1) 必要性:
令

$$A = [\alpha_1, ..., \alpha_n], B = [\beta_1, ..., \beta_n]$$

则 A, B 为正交矩阵, $A^TA = I_n, B^TB = I_n$
令过渡矩阵为 M, 即 $B = AM$, 则 $B^T = M^TA^T$
相乘有 $I_n = B^TB = M^TA^TAM = M^TM$, 所以 M 为正交矩阵.
(2) 充分性:

$$B^TB = M^TA^TAM = M^TM = I_n$$

所以 $\beta_1, \beta_2, ..., \beta_n$ 为标准正交基. □

25.

证. 记方程组为 $Ax = b$,$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$

$$A^TA = \begin{bmatrix} 1 & 2 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 9 & -9 \\ -9 & 18 \end{bmatrix}$$

$$(A^TA)^{-1} = \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 1/9 \end{bmatrix}$$

最小二乘解为

$$x = (A^TA)^{-1}A^Tb = \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 1/9 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 2/9 & 1/9 \\ 1/9 & 1/9 \end{bmatrix} \begin{bmatrix} -9 \\ 27 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

□

26.

解. $[A|b] \to \begin{bmatrix} 1 & 2 & -1 & -1 \\ 2 & 0 & 1 & 1 \\ 2 & -4 & 2 & 1 \\ 4 & 0 & 0 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & -1 \\ 0 & 4 & -3 & -3 \\ 0 & 8 & -4 & -3 \\ 0 & 8 & -4 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & -1 \\ 0 & 4 & -3 & -3 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
$rank(A) < rank(A|b)$, 所以 $Ax = b$ 无解.

$$A^TA = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 2 & 0 & -4 & 0 \\ -1 & 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 1 \\ 2 & -4 & 2 \\ 4 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 25 & -6 & 5 \\ -6 & 20 & -10 \\ 5 & -10 & 6 \end{bmatrix}$$

$$(A^TA)^{-1} = \begin{bmatrix} 5/96 & -7/192 & -5/48 \\ -7/192 & 125/384 & 55/96 \\ -5/48 & 55/96 & 29/24 \end{bmatrix}$$

最小二乘解为

$$x = (A^TA)^{-1}A^Tb = \begin{bmatrix} -11/24 \\ 25/48 \\ 23/12 \end{bmatrix}$$

□

28.

解. $\begin{bmatrix} 2 & 1 & -1 & 1 & -3 \\ 1 & 1 & -1 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 1 & -4 \\ 0 & 1 & -1 & -1 & 5 \end{bmatrix}$
得到解空间的一组基为 $(0,1,1,0,0)^T, (-1,1,0,1,0)^T, (4,-5,0,0,1)^T$
将 $\alpha_1, \alpha_2, \alpha_3$Schmidt 正交化，即令

$$q_1 = \frac{\alpha_1}{||\alpha_1||} = \frac{1}{\sqrt{2}}(0,1,1,0,0)^T$$

$$q_2 = \frac{\alpha_2 - q_1q_1^T\alpha_2}{||\alpha_2 - q_1q_1^T\alpha_2||} = \frac{1}{\sqrt{10}}(-2,1,-1,2,0)^T$$

$$q_3 = \frac{\alpha_3 - q_1q_1^T\alpha_3 - q_2q_2^T\alpha_3}{||\alpha_3 - q_1q_1^T\alpha_3 - q_2q_2^T\alpha_3||} = \frac{1}{3\sqrt{35}}(7,-6,6,13,5)^T$$

□

29.

解. 令 $A = [\alpha_1, \alpha_2]$, 则 $C(A)^\perp = N(A^T) = \{x|A^Tx = 0\}$
得到齐次方程组

$$\begin{cases} x_1 + x_2 + 2x_3 = 0 \\ x_1 + x_3 = 0 \end{cases}$$

得到一个基础解系为 $\alpha_3 = [-1,-1,1]^T$
所以 $C(A)^\perp = span(\alpha_3)$

$$[\alpha_1, \alpha_2, \alpha_3, \alpha] = \begin{bmatrix} 1 & 1 & -1 & 3 \\ 1 & 0 & -1 & 2 \\ 2 & 1 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & -1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 3 & -4 \end{bmatrix}$$

解得唯一解 $(2/3, 1, -4/3)^T$, 所以,$\alpha = \frac{2}{3}\alpha_1 + \alpha_2 - \frac{4}{3}\alpha_3$
其中 $\frac{2}{3}\alpha_1 + \alpha_2 \in C(A)$, $-\frac{4}{3}\alpha_3 \in C(A)^\perp$
所以，α 在 $C(A)$ 中的正交投影为

$$\frac{2}{3}\alpha_1 + \alpha_2 = [5/3, 2/3, 7/3]^T$$

在 $C(A)^\perp$ 中的投影为

$$-\frac{4}{3}\alpha_3 = [4/3, 4/3, -4/3]^T$$

□

31.

解. 令 $\epsilon_1, \epsilon_2, \epsilon_3$ 到 η_1, η_2, η_3 的过渡矩阵为 M.
则

$$[\eta_1, \eta_2, \eta_3] = [\epsilon_1, \epsilon_2, \epsilon_3]M$$

$$M = [\epsilon_1, \epsilon_2, \epsilon_3]^{-1}[\eta_1, \eta_2, \eta_3]$$

$$= \begin{bmatrix} 8 & -16 & 9 \\ -6 & 7 & -3 \\ 7 & -13 & 7 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 3 & 2 \\ -2 & -1 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -3 \\ 1 & 2 & -5 \\ 1 & 3 & -6 \end{bmatrix}$$

所以, T 在 η_1, η_2, η_3 下的表示矩阵为

$$M^{-1}AM = \begin{bmatrix} 1 & 2 & 2 \\ 3 & -1 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

□

33.

解. 令 $\epsilon_1, \epsilon_2, \epsilon_3$ 到 η_1, η_2, η_3 的过渡矩阵为 M.
则

$$[\eta_1, \eta_2, \eta_3] = [\epsilon_1, \epsilon_2, \epsilon_3]M$$

$$T[\epsilon_1, \epsilon_2, \epsilon_3] = [T\epsilon_1, T\epsilon_2, T\epsilon_3] = [\eta_1, \eta_2, \eta_3] = [\epsilon_1, \epsilon_2, \epsilon_3]M$$

所以, T 在 $\epsilon_1, \epsilon_2, \epsilon_3$ 下的表示矩阵为 M.
T 在 η_1, η_2, η_3 下的表示矩阵为 $M' = M^{-1}MM = M$

$$M = [\epsilon_1, \epsilon_2, \epsilon_3]^{-1}[\eta_1, \eta_2, \eta_3]$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & -1 \\ -1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & -3/2 & 3/2 \\ 1 & 3/2 & 3/2 \\ 1 & 1/2 & -5/2 \end{bmatrix}$$

□